Heuristic 1: Defined Data Scan

Scans Ghidra's typed data items once per binary. `s_*` labels yield strings directly; `PTR_s_*` labels are pointer-dereferenced.

From `tailscale/tailscale` — `client/systray.setAppIcon` (stripped build):

```
// Strings:
//   (@\textcolor{phase1col}{\textbf{s\_loading\_00f7a900}}@) = "loading"
```

C pseudocode — recovered string used in a comparison:

```
if ((in_stack_00000008 == s_loading_00f7a900)
    && (DAT_00f7a908 == cStack0000000000000010)) {
  tailscale_com_client_systray_startLoadingAnimation();
}
```

Ghidra typed `s_loading` as string data at `0xf7a900`; the pipeline reads `"loading"` directly from that address.

Heuristic 2: Undefined Data Resolution

Resolves `PTR_s_*` / `s_*` references in C pseudocode absent from the data listing. Parses hex suffix from name; reads Go `(ptr, len)` pairs for precise, length-delimited extraction.

From `usememos/memos` — `plugin/filter.NewAttachmentSchema` (default build):

```
// Strings:
//   (@\textcolor{phase2col}{\textbf{PTR\_s\_attachmentindex\_html\_..\_01aad170}}@) = ["attachment", "memo_id
     "]
//   (@\textcolor{phase2col}{\textbf{PTR\_s\_attachmentindex\_html\_..\_01aacfc0}}@) = ["attachment", "
     filename",
//     "mime_type", "scalar", "string", "attachment", "type",
//     "create_time", "scalar", "timestamp", ...]
```

C pseudocode referencing the symbols (initializing map entries):

```
local_68 = PTR_s_attachmentindex_html_.._01aacfc0;
uStack_60 = _UNK_01aacfc8;
  ...
runtime_mapassign(in_RDI, in_RSI, key, &PTR_DAT_01a76720);
```

Hex suffix `_01aacfc0` parsed → memory at `0x1aacfc0` read as Go `(ptr, len)` pairs → 17 domain-specific strings recovered.

Heuristic 3: DAT Symbols

Resolves `DAT_*` labels (unclassified data). Detects paired string lengths from adjacent stack locals or function arguments.

From `ollama/ollama` — `main.main.func1` in `pull-progress` (default build):

```
// Strings:
//   (@\textcolor{phase3col}{\textbf{DAT\_007ac68b}}@) = "Progress: status=%v, total=%v, completed=%v"
```

C pseudocode — used as a format string argument:

```
w.tab = (internal_abi_ITab *)0x2c;   // length = 44
format.len = (int)&DAT_007ac68b;
format.str = extraout_RDX;
fmt_Fprintf(w, format, in_stack, 3, ...);
```

Hex address parsed from label. Length `0x2c` (44) detected from adjacent assignment, matching the 44-byte format string.

Phase 4: Hex Literals

Resolves bare hex constants in C pseudocode that fall within `.rodata` address ranges.

From `argoproj/argo-workflows` — `v1alpha1.ClusterWorkflowTemplate.GetResourceScope` (stripped build):

```
// Strings:
//   (@\textcolor{phase4col}{\textbf{0x3171032}}@) = "cluster"
```

C pseudocode — bare hex address returned as a string pointer:

```
undefined8 ...GetResourceScope(void) {
  return 0x3171032;
}
```

Address `0x3171032` falls within `.rodata`. The decompiler shows only a numeric constant; Phase 4 resolves it to `"cluster"`.